高压电缆逸出气体的三域指纹特征及状态诊断

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摘要：高压电缆的逸出气体可来自交联残留物的释放，也可由受潮腐蚀、材料热降解或放电产生。由于同一种气体可能来自不同过程，仅凭该气体的浓度难以判定其来源。本文提出由氢气浓度域（C 域）、热产气组成域（T 域）和放电相关信号域（D 域）构成的三域指纹方法。C 域依据 H_2 浓度分档。T 域依据 C_2H_4和 C_2H_6浓度判定是否启用，并依据两者的浓度比或乙烷报告值为 D 零时的专用规则确定编码。D 域依据 C_2H_2浓度判定是否启用，结合 C_2H_2/C_2H_4确定编码，并设置乙炔浓度超过 500 ppm 时的升级条件。CH_4用于背景释放核查，CO 和 CO_2用于构建辅助氧化子指纹。选取 4 条线路的 10 条现场本体测试记录，复算各域编码和综合警示等级，并结合解剖照片进行外观对照。结果表明，L05 线路 A、B 两相均为轻微警示，A 相仅触发 T 域，B 相同时触发 C 域和 T 域。L08 线路 A 相的乙炔浓度为 680 ppm，使 D 域编码由按浓度比判定的 D1 升至 D2，综合警示等级仍为 3 级。结合各域编码和触发条件，可区分同一综合警示等级下的气体特征，为确定现场复核方向及优先级提供依据。警示等级与实体损伤程度之间的对应关系仍需独立检验。

关键词：高压电缆；逸出气体分析；三域指纹；缓冲层；状态警示

Three-Domain Fingerprint Characteristics and Condition Diagnosis of Outgassed Species from High-Voltage Cables

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Abstract: Crosslinking residues in high-voltage cable materials can release gases. Moisture-induced corrosion, thermal degradation, and electrical discharge can also generate gases. Because these sources can yield overlapping gas species, the concentration of a single species is insufficient to distinguish among them. This study proposes a three-domain fingerprinting method comprising a hydrogen concentration domain (C), a thermal gas composition domain (T), and a discharge-related signal domain (D). The C-domain code is assigned by H_2 concentration range. The T domain is activated based on C_2H_4 and C_2H_6 concentrations. Its code is determined by the C_2H_4/C_2H_6 concentration ratio or by a separate rule when the reported C_2H_6 concentration is zero. The D domain is activated based on the C_2H_2 concentration. Its code is determined using the C_2H_2/C_2H_4concentration ratio, with an override when the C_2H_2 concentration exceeds 500 ppm. CH_4 is used to check for possible background gas release, and CO and CO_2 are used to construct an auxiliary oxidation fingerprint. The domain codes and overall alert levels were recalculated for ten field test records from cable bodies on four circuits. The results were compared with visible features in photographs taken during cable dissection. On circuit L05, phases A and B both received a slight alert. Phase A activated only the T domain, whereas phase B activated both the C and T domains. On circuit L08, the C_2H_2 concentration in phase A was 680 ppm, which raised the D-domain code from D1, as determined by the concentration ratio, to D2. The overall alert remained at level 3. Considering the domain codes together with their triggering conditions distinguishes gas profiles assigned the same overall alert level and helps identify which field checks to perform and in what order. The relationship between alert level and physical damage severity remains to be independently evaluated.

Key words: high-voltage cable; evolved gas analysis; three-domain fingerprint; buffer layer; condition alerting